A Label-Free Runtime Safety Monitor for Frozen End-to-End Driving Planners,
Evaluated Closed-Loop at Benchmark Scale

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Abstract

End-to-end driving planners fail catastrophically in safety-critical closed-loop scenarios — published NeuroNCAP results score UniAD at 1.84/5 with collisions in 88–98% of runs — yet the field's dominant open-loop metrics cannot see it. We build a runtime monitor that reads only a frozen planner's own outputs (its plan, detected objects, and their tracked motion; no labels, no training, no privileged simulator state) and intervenes with a latched, threat-cleared stop. Across seventeen pre-registered iterations and an independent verification pass we show: (1) the published NeuroNCAP UniAD baseline *independently reproduces* (pooled 2.12 vs. 1.84 at 20 seed-paired runs per scenario), to our knowledge a first; (2) a union of two label-free geometric detectors with a threat-cleared release lifts the full-benchmark score to **2.91** (+0.783, 95% CI [+0.605, +0.928]) while leaving clean-scene behaviour identical to the unmonitored planner, at a deployment-metric cost that is a tight null (safety×progress delta −0.03, 95% CI [−0.13, +0.07]); (3) the committed stop is a *position guarantee* that no tested softer intervention replaces — three evasive maneuvers, a calibrated crawl, and a geometric threat-router were all refuted by pre-registered falsifiers, the router while demonstrating that a deployment-positive monitor is achievable (+0.226 vs. the unmonitored planner, CI [+0.004, +0.421], voided by its safety gate); (4) an RSS-style formal envelope on identical inputs achieves the campaign's best raw safety *by near-paralysis*, quantifying closed-loop the over-conservatism the literature only asserts; (5) the planner's own candidate trajectories *collapse under threat* on two planners (UniAD: 14 m benign diversity to 4 cm; VAD: partial, below a pre-registered viability bar) — the first threat-conditioned diversity measurements on end-to-end planners' own candidates; and (6) two independent failure analyses — cross-planner selectivity transfer and routing safety — converge on a single binding constraint: *tracking quality.* One headline claim was withdrawn by our own audit and re-established on independent data; the withdrawal, six published nulls, and the raw per-frame evidence for every number are part of the public record.

1 Introduction

Open-loop planning metrics on nuScenes are saturated and gameable: an ego-state MLP with no perception outperforms published end-to-end planners on displacement error [2]. Closed-loop evaluation under safety-critical perturbation tells the opposite story — state-of-the-art planners collide in the vast majority of adversarial episodes [1]. This gap defines the honest axis for driving-safety research, and it is where runtime assurance belongs: if a frozen planner's failures are predictable from its own outputs, a small monitor can intervene before they become collisions, without retraining, fleet data, or privileged state.

This paper reports a complete measurement campaign on that thesis. Its contributions are the six numbered results in the abstract; its method is as much a contribution as its numbers: every iteration was pre-registered with falsifiers before data; every comparison is seed-paired on deterministic episodes; every number regenerates from committed raw evidence; and the campaign's one statistical over-claim was caught by its own audit, withdrawn in place, and re-established on fresh data.

2 Related work

Runtime monitors for end-to-end planners. CATPlan/RiskMonitor [3] is the canonical introspection-style monitor: a *learned* transformer decoder trained on the sign of the planner's collision loss, reading UniAD/VAD internal queries, with a braking policy (66.5% closed-loop collision-rate reduction on NeuroNCAP). Argus [4] attaches a takeover fallback that generates its own replacement trajectory in CARLA. Centaur [5] performs test-time training. Our monitor differs on each axis: label-free geometry over the planner's own outputs (no trained head), no invented takeover trajectory (the campaign's evasion, crawl, and routing nulls argue this is not merely simpler but structurally safer), no weight updates — and it is validated with a progress-aware metric none of the above report.

Runtime selection among a frozen planner's candidates. Candidate scoring is an active paradigm [6, 7, 8], but in every verified instance the scorer is co-trained within the planner's pipeline and evaluated on non-reactive NAVSIM. No verified published work re-ranks a *frozen* planner's native candidates at runtime with an external safety signal — the mechanism our measurements close for command-indexed candidates (the candidates collapse when needed) and reopen for diversity-trained heads.

Mode collapse under threat is documented as a training-time phenomenon [9], with explicit metrics only prediction-side [10]. Our threat-conditioned diversity measurements on planners' own candidate sets, closed-loop, on two planners, appear to be the first. **Deployment-aware statistics** (progress+safety with bootstrap CIs, intervention budgets, detection lead time) and a **closed-loop quantification of formal-envelope over-conservatism** [11] likewise appear unreported in the verified corpus.

3 Apparatus and methodology

Three public containers on one L4 GPU: the NeuroNCAP orchestrator (scenario actor and scorer), the NeuRAD neural renderer (photoreal multi-camera frames from real nuScenes drives), and the frozen planner serving `/infer` carrying the monitor as a ∼150-line patch, env-gated per arm. NeuroNCAP episodes replay deterministically per run index — a property we established mid-campaign, which first invalidated an early pooled claim (§9) and then upgraded every comparison to seed-paired on identical episodes. The determinism is verified at scale: the 20-run measurement's first six indices reproduce the earlier 6-run measurement exactly, on every scenario pair, in both arms.

The methodological spine, held for all seventeen iterations: hypotheses and falsifiers frozen in writing before data; seed-paired within-pair bootstrap confidence intervals on every delta; per-frame monitor decision logs, per-run trajectories, and run logs committed for every arm; nulls published with the same weight as wins.

4 The monitor

The union brakes when (plan-vs-tracked-path closest approach < 1.5 m) OR (observed agent-closing time-to-collision < 2.5 s). The two terms correspond to physically distinct failure modes: the first catches the side crossing, the second the head-on that the planner's own optimism hides (its plan claims 3–4 m of clearance on runs that end in collision). Object velocity is *observed* — ego-motion-compensated tracking by persistent identity — not the planner's forecast; the forecast is optimistic on exactly the runs that collide (the planner's own outputs predict its collisions at AUROC 0.83). The stop is latched — safe even when the trigger is wrong — and releases after four consecutive verified-clear frames, returning control; the release strictly dominates the plain latch (identical safety, safe-progress +0.246, CI [+0.206, +0.293]).

5 The benchmark result

pooled, 14 official scenes (n=20/pair)	unmonitored UniAD	+ released union
NeuroNCAP score (0–5)	2.12 (published 1.84)	**2.91**
side collisions	74%	**44%**
stationary collisions	29%	**18%**
frontal (score / collisions)	1.24 / 78%	1.78 / 90%
safe-progress	2.40	2.36

Table 1: The power measurement: 799 episodes, seed-paired. Benchmark delta +0.783, CI [+0.605, +0.928]; deployment delta −0.03, CI [−0.13, +0.07].

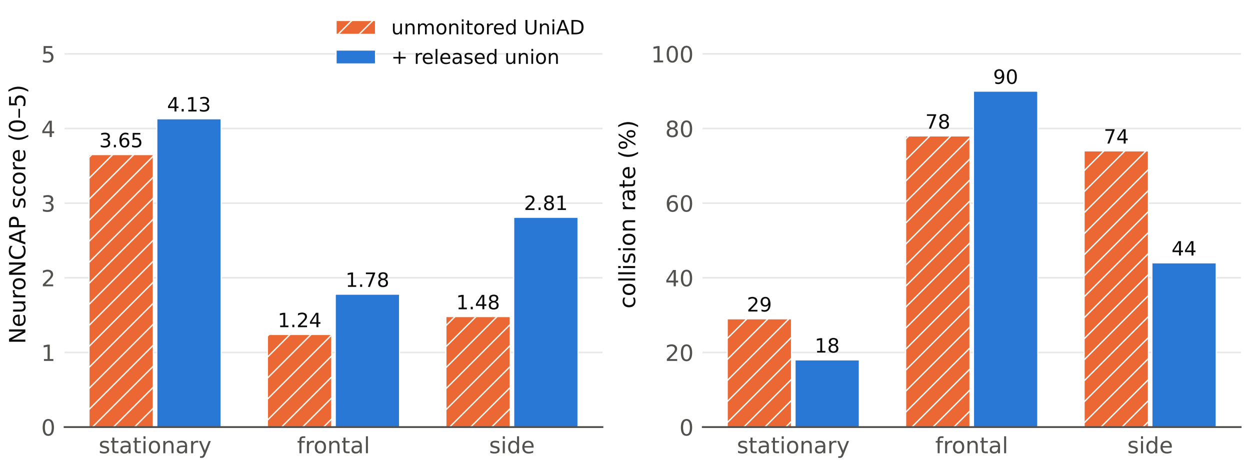

Figure 1: Per-class results at n=20 runs per pair. Frontal remains mitigation, not prevention; the frontal/0346 regression is confirmed real and reported as the stop policy's named cost.

In safety-case units (from committed decision logs): median detection lead 2.5 s before counterfactual contact (Fig. 2); 11 brake frames per 242 benign meters; frontal mean impact speed 13.9 → 6.7 m/s.

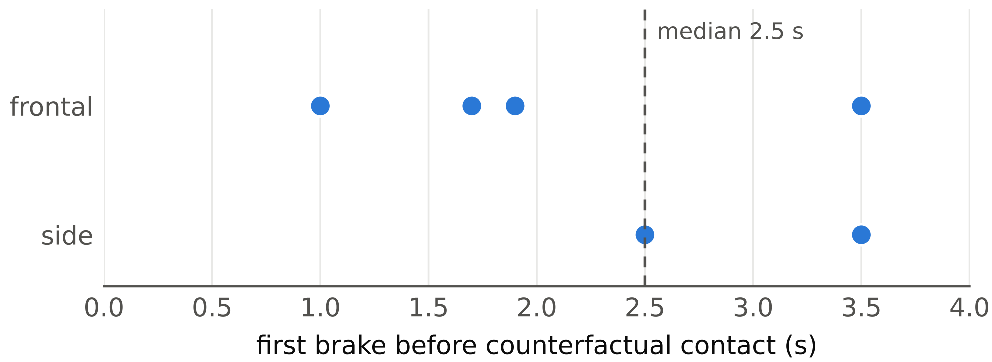

Figure 2: Detection lead time: the six lead-time events measurable under the strict contact criterion, shown as events (a histogram of six values would over-claim).

6 Negative results I: the stop is a position guarantee

Five designs to keep more progress than the stop were pre-registered and refuted, and the pattern of their failures is the finding: three evasive maneuvers fail under *false* alarms (the third crashes the benign scene at 25% vs. the unmonitored 10%); a calibrated 2.0 m/s crawl fails under *true* alarms (side 37% → 57%: it delivers the ego into the crossing point the stop halts short of); and a geometric threat-router (stop on projected path overlap with the planned corridor; crawl otherwise — Fig. 3) comes closest, recording the campaign's first deployment CI excluding zero against the unmonitored planner (+0.226, CI $[+0.004, +0.421]$) while still failing its pre-registered gate on a single misrouted crossing. The stop is not merely speed reduction but a position guarantee — the only tested intervention safe in both error directions — and a deployment-positive monitor is demonstrably achievable, pending a crossing-safe routing signal (§8).

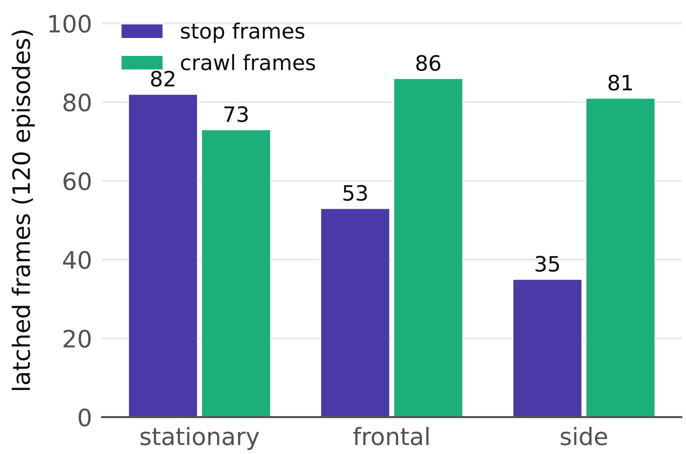

Figure 3: The router's latched frames by action and class: the crawl share is where its deployment gain lives; its side-class misroute is where the gain died.

7 Negative results II: no plan B, envelope paralysis

The planner has no plan B. Pre-registered checkpoints measured whether a low-risk candidate exists when the executed plan is dangerous: UniAD's command-conditioned candidates span 13.9 m in benign frames and collapse to a 4 cm spread under threat (0/37 escapes); VAD's native modes retain partial diversity (21% escapes) but stay below the frozen 30% viability bar. A diversity-trained candidate head is the motivated successor mechanism.

Stopping power is free; selectivity is not. An RSS-style guaranteed-stopping envelope on identical inputs and actuator posts the campaign's best raw safety (clean 0%, frontal 30%, side 0%) by driving 3.6–8.2 m where the planner drives 21–32 m — safe-progress 0.88 against the unmonitored 1.83 (union − RSS = +1.345, CI $[+0.944, +1.701]$).

8 The binding constraint is tracking quality

Two failure analyses, from independent directions, converge. *Transfer:* on a frozen VAD the union prevents exactly the failures VAD has (stationary 85% → 0%, side 65% → 0%) but loses selectivity everywhere; decision logs attribute the over-braking to nearest-neighbor identity jitter manufacturing closing speed. *Routing:* the router's single misrouted crossing traces to velocity dropout across identity switches; all three named per-frame geometric repairs were refuted offline on the committed logs (a provable no-op; a dead trade; a non-separable feature). The discriminating signal in both cases is velocity continuity through identity switches — a property of the tracking layer, not of any decision rule above it. An association-and-filter layer built to this specification converts 12 of 13 unsafe crawl frames offline yet fails its pre-registered every-frame bar by a single borderline frame (0.2 m) — the gate held, and the closed loop stayed off; the result and its open-loop-replay fidelity caveat are part of the record.

9 Verification, reproducibility, and the incident record

An independent verification pass re-derived every claim from raw evidence. It discovered the episode determinism, found that an early pooled validation had counted deterministic replays as independent replications, withdrew the headline, and re-measured on genuinely unique episodes where the claim was re-established (mini-scene safe-progress +0.398, CI $[+0.133, +0.665]$). The power measurement's incident record is reported as part of the measurement: five machine-freezing events across two physical hosts, root-caused via an on-box vitals instrument to memory exhaustion on a swapless image; one scenario pair reported at n=19 after its final episode reproducibly froze the pre-swap host. No completed episode was lost or re-measured differently — the exact-reproduction gate is the proof. Every number regenerates from committed raw evidence with the commands in the repository README.

10 Limitations

One simulator (NeuroNCAP/NeuRAD), whose deterministic replay we characterize and exploit; 20 runs per pair against the published 100-run protocol; two planners; the VAD transfer bounded by an identity-association layer's quality; the RSS baseline is the longitudinal form; no fleet data or real-vehicle validation. Frontal head-on prevention (as opposed to mitigation) remains open and is, on this evidence, not reachable by maneuver invention from a label-free monitor.

11 Conclusion

A label-free monitor on a frozen planner lifts the full official NeuroNCAP benchmark from an independently reproduced 2.12 to 2.91 with a confidence interval excluding zero, at a deployment-metric cost statistically indistinguishable from zero. The campaign's negative results are as load-bearing as its wins: the stop's position guarantee, the planner's missing plan B, the envelope's paralysis, the achievable-but-not-yet-safe deployment flip, and the convergence of two independent failure modes on tracking quality — together they map the mechanism space for runtime driving assurance more sharply than the headline number does, and they name the next frontier precisely.

References

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